PharmAchieve

ACUTE CORONARY SYNDROME & STABLE ANGINA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ACS	Acute Coronary Syndrome
ADR	Adverse Drug Reaction
BB	Beta Blocker
BMS	Bare Metal Stent
CABG	Coronary Artery Bypass Graft
CAD	Coronary Artery Disease
CCB	Calcium Channel Blocker
CHD	Coronary Heart Disease
CHF	Congestive Heart Failure
C/I	Contraindication
CK-MB	Creatine Kinase Myocardial Band
COX	Cyclooxygenase
CRP	C-Reactive Protein
CVD	Cardiovascular Disease
DAPT	Dual Antiplatelet Therapy
DES	Drug-Eluting Stent
DM	Diabetes Mellitus
EF	Ejection Fraction
HCTZ	Hydrochlorothiazide
HF	Heart Failure
HRT	Hormone Replacement Therapy
HTN	Hypertension
IHD	Ischemic Heart Disease
ISDN	Isosorbide Dinitrate
LA	Long Acting
LDL	Low-Density Lipoprotein
LMWH	Low Molecular Weight Heparin
LVEF	Left Ventricular Ejection Fraction
MI	Myocardial Infarction
NSAID	Nonsteroidal Anti-inflammatory Drug
NSTEMI	Non-ST Elevation Myocardial Infarction
NTG	Nitroglycerin
PAD	Peripheral Artery Disease
PCI	Percutaneous Coronary Intervention
STEMI	ST Elevation Myocardial Infarction
TIA	Transient Ischemic Attack
UA	Unstable Angina
UFH	Unfractionated Heparin

DEFINITIONS

Atherosclerotic CVD

Clinical manifestations of atherosclerosis (deposition of fatty materials/plaques within the arterial walls resulting in thickening and reduced blood flow)

- **Stroke**
  Inadequate oxygen supply to areas of the brain
- **CAD/CHD/IHD**
  Inadequate oxygen supply to myocardium
  - **Stable angina**
    Chest pain due to reduced oxygen supply to myocardium
  - **Acute Coronary Syndrome**
    Unstable angina, NSTEMI, and STEMI
- **Peripheral Artery Disease (PAD)**
  Inadequate oxygen supply to lower extremities

PATHOPHYSIOLOGY OF ATHEROSCLEROSIS

Excessive fat and LDL accumulate within the circulation

↓

Accumulated LDL induces innate immune response to engulf LDL particles, leading to foam cell formation

↓

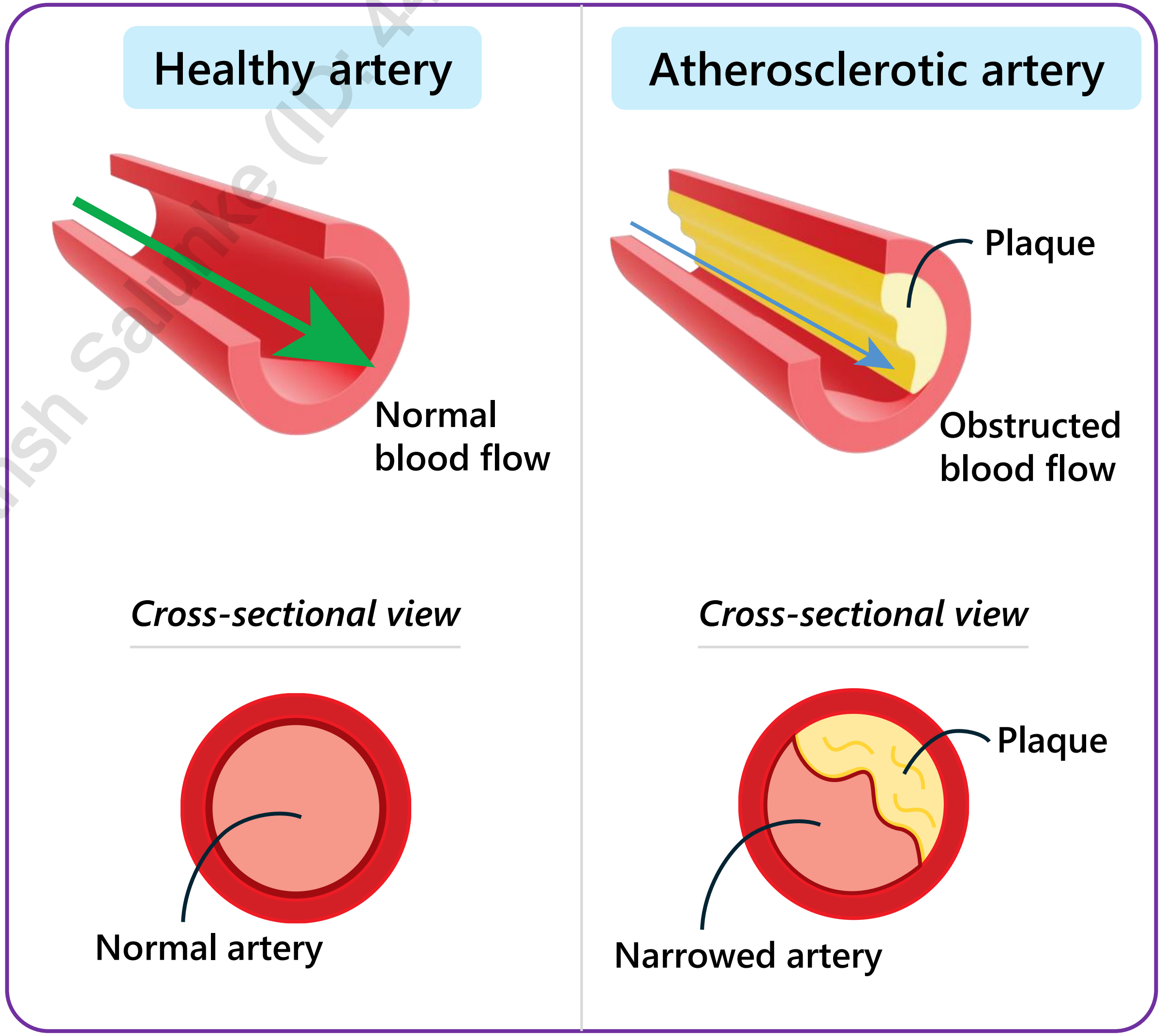
Foam cells lead to formation of fatty streak/plaque accumulation in the walls of arteries

↓

Atherosclerotic CVD (Stroke, CAD, ACS)

↓

Death



CLINICAL PRESENTATION

Acute Coronary Syndromes (i.e. STEMI, NSTEMI, Unstable Angina)	Stable Angina
• Chest heaviness, pressure, discomfort • Pain radiating to the jaw/shoulder/arm • Shortness of breath (SOB) • Dull, sub-sternal pain • Fainting, sweating • Nausea/vomiting • Lasts for longer periods of time (> 5 minutes) • Not relieved by nitroglycerin • Unpredictable onset (occurs at rest)	• Chest heaviness, pressure, pain, discomfort • Chest symptoms may radiate to upper body • Lasts for ≤ 5 minutes • Relieved by rest or nitroglycerin • Predictable onset (with strenuous activity)

CONTRIBUTORY FACTORS FOR ACS

Drug Causes

- NSAIDs, including COX-2 inhibitors
- Stimulants: amphetamines, sympathomimetic agents (cocaine)
- Oral contraceptives
- Alcohol

Aggravating Factors

- Cold exposure, physical exertion, emotional distress, anxiety, hypotension, hypertension, arrhythmias

Risk Factors

- CVD (e.g. HTN, dyslipidemia), smoking, diabetes mellitus, chronic kidney disease, obesity, physical inactivity, age, male sex
- Family history: 1st degree relative with MI (men age $<$ 55 years, women age $<$ 65 years)

CARDIAC FUNCTION BIOMARKERS

FYI

Troponin (preferred)

- A highly accurate, sensitive and specific indicator of myocardial cell injury
- Protein released from myocytes when myocardial damage occurs
- Not a specific marker for plaque rupture, erosion, and/or ulceration and thrombosis precipitating ACS
- Time to initial rise: 1-3 hours with highly sensitive troponin assays (preferred)
- Peak: 18-24 hrs
- Time to return to normal: up to 14 days

Creatine Kinase

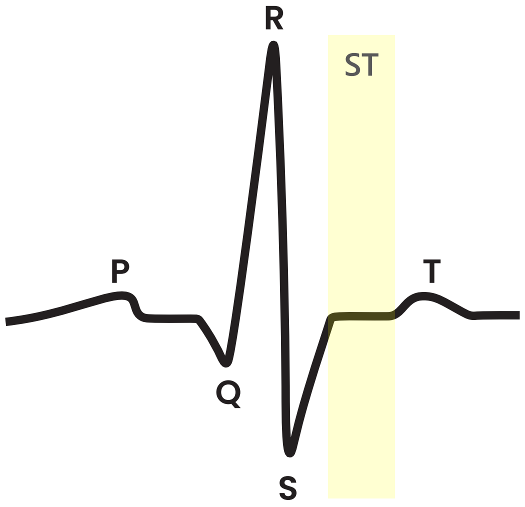
- Myocardial muscle creatine kinase (CK-MB) found mainly in the heart
- Time to initial rise: 3-12 hrs
- Peak: around 24 hrs
- Time to return to normal: 48-72 hrs

MYOCARDIAL INFARCTION ON ECG

MYOCARDIAL INFARCTION (MI): STEMI & NSTEMI

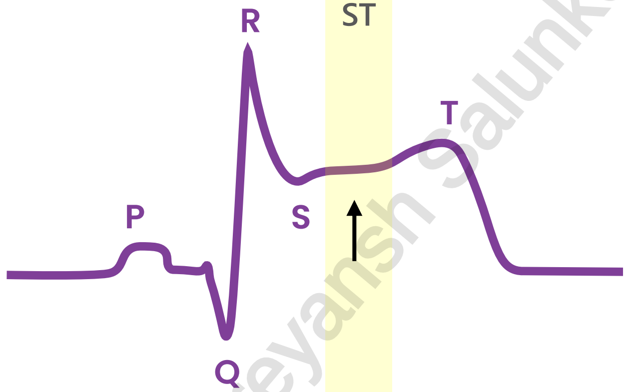
* May also show a T-wave inversion or even a normal ECG

Normal



ST segment – refers to plateau phase of ventricular transmembrane action potential

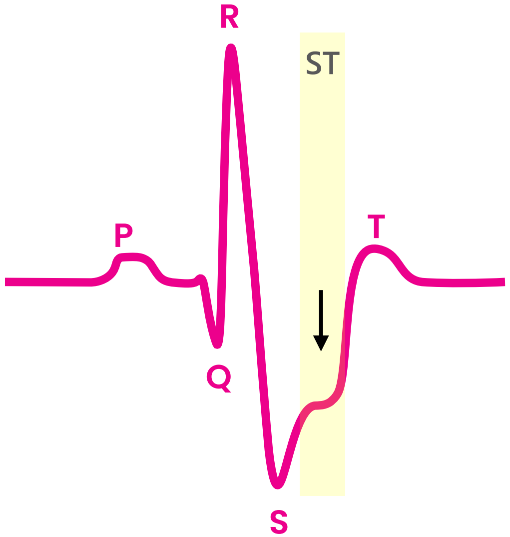
STEMI



ST **elevation** + elevated troponin

Complete occlusion of blood vessels resulting in necrosis of myocardium

NSTEMI



ST **depression*** + elevated troponin

Partial occlusion of blood vessels resulting in necrosis of myocardium

Treat immediately to reduce risk of complications (↓ ventricular function and cardiac output)

TIMI RISK SCORE

THROMBOLYSIS IN MYOCARDIAL INFARCTION

→ *A risk stratification tool used for patients with unstable angina or NSTEMI*
→ *Estimates risk of mortality, MI, or recurrence of ischemic events, thereby guiding therapeutic decision making*

1 point is assigned for each of the following:

- Age ≥ 65 years
- ≥ 3 CVD risk factors *(dyslipidemia, HTN, DM, current smoking, family history of CAD)*
- ≥ 50% coronary artery stenosis
- Any ASA use within the past 7 days
- ≥ 2 episodes of severe angina within the last 24 hrs
- Elevation in cardiac markers *(troponin or creatine kinase)*
- ST segment deviation ≥ 0.5 mm on ECG

Interpretation of TIMI score:

- A higher score increases the risk of cardiac events
  - *TIMI 0/1 = 5% risk*
  - *TIMI 6/7 = 41% risk*
- Low risk: 0-2
- Intermediate risk: 3-5
- High risk: 6-7

ANGINA

SYMPTOMS AND CLASSIFICATION

Symptoms: chest pain, nausea, abdominal pain, sweating, anxiety, tiredness, and shortness of breath.
* Women, the elderly and patients with diabetes may present with atypical symptoms (e.g. dyspepsia)

Chronic Stable Angina (Angina Pectoris)

- Exercise-induced
- No/minimal ECG changes with no changes in troponin

Grade I	• No symptoms with ordinary physical activity • Symptoms with strenuous, rapid or prolonged activity
Grade II	• Slight limitation with ordinary physical activity
Grade III	• Marked limitation with ordinary physical activity
Grade IV	• Unable to continue activity without discomfort, may have symptoms at rest

Unstable Angina

- Often occurs while resting, sleeping or with minimal physical activity
- ST depression* with no changes in troponin
- Can lead to a heart attack

* May also show a T-wave inversion or even a normal ECG

GOALS OF THERAPY

ACS

- ✓ Restore blood flow to the infarcted artery (STEMI)
- ✓ Prevent total occlusion of the artery (NSTEMI/UA)
- ✓ Prevent re-occlusion
- ✓ Reduce or contain the size of the infarct
- ✓ Prevent death or complications
- ✓ Relieve symptoms of ischemia
- ✓ Minimize ADR of medications

Stable Angina

- ✓ Relieve symptoms of acute ischemia
- ✓ Improve exercise tolerance and quality of life
- ✓ Decrease frequency and severity of repeat ischemic episodes
- ✓ Prevent ACS and death
- ✓ Treat modifiable risk factors for CAD
- ✓ Minimize ADR of medications

STABLE ANGINA PECTORIS

TREATMENT ALGORITHM

Angina Pectoris

- **Improve quality of life (Symptom Management)**
  - **Acute treatment**
    - * NTG SL tabs 0.3-0.6mg Q5min* PRN
    - * NTG spray 0.4mg Q5min* PRN
  - **Chronic treatment**
    - BB **
    - LA-Nitrate (when BB or LA-CCB cannot be used)**
    - LA-CCB ** (non-DHP used only in normal LVEF)
- **↓ mortality**
  - <u>Start</u>:
    - Antiplatelet (ASA 81 mg daily, clopidogrel 75 mg daily if intolerant to ASA) ***
    - High-intensity statin to target LDL <1.8 mmol/L or ≥ 50% reduction
    - ACEi or ARB
    - BB (mortality benefit in certain populations)
  - <u>Stop</u>:
    - HRT
    - NSAIDs

* If discomfort persists for 5 min after 1 dose, call ambulance. Administer up to 3 doses until EMS arrives)

** Use combination therapy if symptoms are not controlled with monotherapy; use BB as initial therapy

*** COMPASS trial found ASA 100 mg daily + rivaroxaban 2.5 mg BID reduced CV events/death more than ASA alone, but bleeding was increased → treatment must be individualized to determine risk vs. benefit

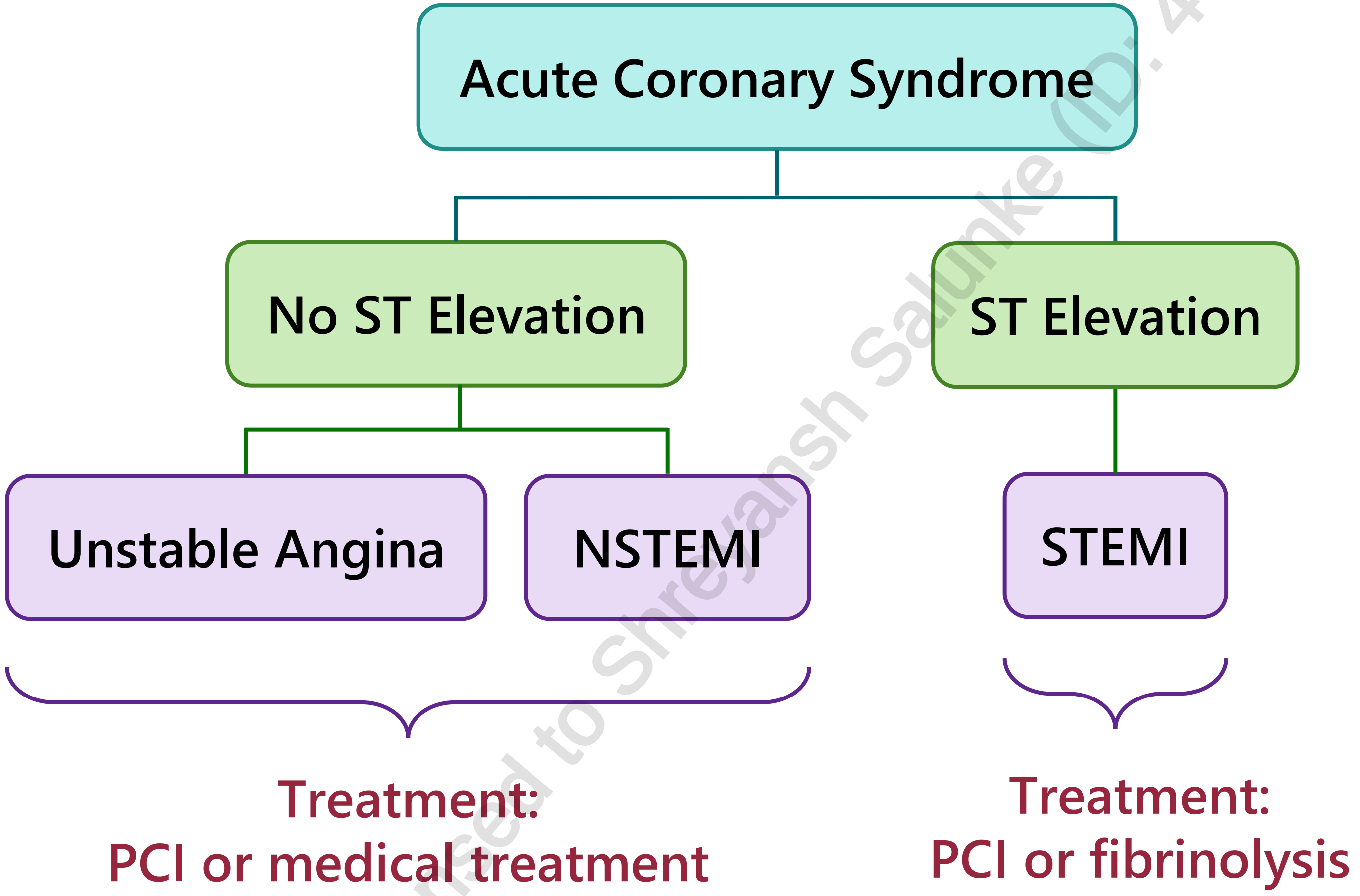
STABLE ANGINA

- Beta-blocker often preferred first-line for management of chronic SA in patients with previous MI or CHF (more benefit in reduced EF)
- Otherwise, patient can be treated with either beta blocker or CCB as first-line treatment for uncomplicated patients

- If suboptimal relief of symptoms, may consider:
  - Switching to another therapy
    OR
  - Combining therapies
    - Caution with beta-blocker and non-DHP CCB due to risk of severe bradycardia, AV block, excessive fatigue

- If symptoms are not relieved despite maximally tolerated combination of medications, consider revascularization

TREATMENT - UA/NSTEMI VS STEMI



NSTEMI AND UNSTABLE ANGINA

Troponin, ECG, CBC, electrolytes, glucose, Cr
- ASA 160-325 mg STAT
- Nitroglycerin IV or SL (unless SBP<90)
- Beta Blocker (if CI: CCB)
- Heparin (UFH IV, LMWH SC, or Fondaparinux SC)
- P2Y12 inhibitor (clopidogrel or ticagrelor)
- O_2 PRN (maintain O_2 >90%)
- Morphine PRN (analgesic)

High risk patient (at least 1 of the following)?
+ve cardiac enzymes, ST-segment changes, TIMI 6-7, recurrent ACS, prior PCI/CABG, HF, hemodynamic instability, ventricular tachyarrhythmia

Yes

Consider PCI +/- Glycoprotein IIb/IIIa inhibitor (angioplasty)
Optimize treatment (ASA and antianginals)

No

Medical management:
Optimize antianginals
Continue ASA, heparin, P2Y12 inhibitor, BB
Start ACEi within first 24 hours post-MI
Start high-potency statin

STEMI

- ASA 160-325 mg STAT
- NTG (IV if SL ineffective)
- BB (once hemodynamically stable)
- P2Y12 inhibitor (loading dose)
- Heparin (UFH IV, LMWH SC, or Fondaparinux)
- PRN: Morphine, O_2

Can PCI be done in ≤ 120 minutes (ideally ≤ 90 minutes)?

Yes → PCI

No → Contraindications to fibrinolysis? †

Yes → PCI

No → Fibrinolysis (with alteplase or Tenecteplase) door to needle ≤ 30 minutes → Pharmacoinvasive Strategy *

Note: Reperfusion within 12 hours of symptom onset has a significant mortality reduction benefit

- BB and ACEi (start within 24 hours)
- ASA 81 mg daily (continue indefinitely unless contraindicated)
- P2Y12 inhibitor – DAPT X 1 yr → ticagrelor is usually preferred, but clopidogrel is the drug of choice in fibrinolysis
- Heparin (UFH IV, LMWH SC, or Fondaparinux) – continued for at least 48 hrs, or until PCI is underway
- High-intensity statin

† Contraindications to fibrinolysis:
- Active or prior ICH
- Suspicion of aortic dissection
- Intracranial neoplasm
- Cerebrovascular lesions
- Severe hypertension
- Active bleeding
- Current anticoagulant use
- Serious head trauma or stroke within the last 3 months

* Pharmacoinvasive strategy = routine angiography after fibrinolysis +/- PCI, or PCI for patients with ineffective fibrinolysis, and routine transfer to a PCI site if applicable

POST STEMI/NSTEMI MANAGEMENT

Non-pharmacological measures	• Rehabilitation • Weight management • Exercise programs • Psychosocial issues (e.g. stress management, etc.)

Pharmacological measures	• Dual antiplatelet therapy (DAPT): ASA + P2Y12 inhibitor po daily
	  • Following revascularisation (PCI or CABG): ticagrelor or prasugrel
	  • Following medically managed ACS without revascularisation:
	    • *Ticagrelor > Clopidogrel > Prasugrel
	      • *Weak recommendation with low quality evidence to suggest choosing ticagrelor over clopidogrel
	• High-intensity statin
	• Beta blocker
	• ACEi

PHARMACOLOGICAL MEASURES - ANTICOAGULANTS

	UFH	Enoxaparin	Fondaparinux	Bivalirudin
STEMI + PCI	Treatment of choice	Alternative	NOT preferred (increased risk of catheter-related thrombosis)	Alternative
STEMI + fibrinolysis	Alternative	Treatment of choice	Alternative	NOT preferred (used in patients undergoing PCI only)
NSTEMI/UA	Alternative	Treatment of choice	Treatment of choice	Alternative if undergoing PCI
Severe renal impairment (CrCl < 30 mL/min)	No adjustment required	Reduce dose	Avoid	Adjust infusion rate
Administration	IV	SC	SC	IV
Other	Frequent lab monitoring, reversal agent available (protamine), risk of heparin induced thrombocytopenia (HIT)	Reduce dose if patient is >75 years old	Less bleeding	Expensive

PHARMACOLOGICAL MEASURES - ANTIPLATELETS

Glycoprotein IIb/IIIa Inhibitors	• E.g. eptifibatide • Use is mainly for patients undergoing PCI • Can also be used for persistent ischemia despite optimal DAPT
ASA	• Low dose ASA continued indefinitely, unless significant bleeding risk or on an anticoagulant (see atrial fibrillation lecture)

DAPT- P2Y12 Inhibitors	• Clopidogrel
	    • First-line for STEMI patients undergoing fibrinolytic therapy (in combo with ASA)
	• Ticagrelor
	    • Preferred over clopidogrel in ACS without PCI
	• Prasugrel
	    • Rarely used

PHARMACOLOGICAL MEASURES - ANTIPLATELETS

	Clopidogrel (C)	Ticagrelor (T)	Prasugrel (P)
Potency of P2Y12 inhibition	T, P > C		
Mechanism of inhibition	Irreversible	Reversible	Irreversible
Metabolism	Prodrug CYP2C19	Active moiety CYP3A4	Prodrug CYP3A4, CYP2B6
Efficacy	T, P > C		
Bleeding	C < T, P		
Adverse effects/precautions	Rash	Dyspnea	CI: history of stroke, < 60 kg or age > 75
Drug Interactions	CYP2C19 inhibitors (e.g. omeprazole, esomeprazole	CYP3A4 inhibitors (e.g. clarithromycin)	NSAIDs (applies to all antiplatelets)
Dosing	Loading: 300 – 600 mg Maintenance: 75 mg daily	Loading: 180 mg Maintenance: 90 mg BID	Loading: 60 mg Maintenance: 10 mg daily
Min. # of days to discontinue before CABG	Semi-urgent: 2-3 days Elective: 5 days	Semi-urgent: 2-3 days Elective: 5 days	Semi-urgent: 5 days Elective: 7 days

DURATION OF DAPT

- Standard 1-year DAPT: ASA + P2Y12 inhibitor (ticagrelor or prasugrel > clopidogrel) then stop P2Y12, continue ASA indefinitely
- Minimum duration of 2nd antiplatelet drug depends on diagnosis and the type of stent:

Patient scenario	Duration of DAPT
PCI with BMS (bare metal stent)	1 year (minimum 30 days)
PCI with DES (drug eluting stents) • DES are newer technology, shown to reduce risk of clot forming on the stent, leading to stent thrombosis	1 year (minimum 3-6 months)
STEMI with fibrinolytic therapy	1 year (at least 14 days based on evidence)
NSTEMI with no PCI	1 year
High bleeding risk with PCI	Minimum 1-3 months

1 YEAR!!!

After 1 year, determine bleeding risk

- If not at high risk of bleeding, continue DAPT for up to 3 years (If on ticagrelor, reduce dose to 60 mg BID)

PHARMACOLOGICAL MEASURES

ACEi/ARB

- Initiate within 24 hours of ACS and titrate to target doses
- Improves HF, reduces mortality, and reduces the likelihood of recurrent ACS
- ARB does not have the same evidence and is only indicated when ACEi is not tolerated
  - Valsartan is the only ARB studied post-MI that had similar efficacy to ACEi
- Indicated for all patients post-MI and all patients with stable angina (without contraindications)

Statins

- Initiate <u>high intensity</u> treatment
- High potency: rosuvastatin, atorvastatin (most commonly seen post-ACS)
- Medium potency: simvastatin, pravastatin, lovastatin
- Low potency: fluvastatin
- Indicated for all patients post-ACS and all patients with stable angina

PHARMACOLOGICAL MEASURES

Beta-blockers (e.g. metoprolol, atenolol)

- Stable angina: all BBs are equally effective – preferred first-line for select patients
- ACS: start when patient is hemodynamically stable, has mortality benefits and is recommended in all patients post-MI (unless contraindicated)
- Contraindications: reactive airway disease, bradycardia (heart rate ≤50 bpm), second- or third-degree heart block without a functioning pacemaker, hypotension (systolic blood pressure [SBP] <100 mm Hg)
- Do not abruptly discontinue BBs → can cause rebound tachycardia or exacerbation of angina

Calcium Channel Blockers (CCB)

- Non-DHP → diltiazem, verapamil
- DHP → amlodipine, nifedipine ER
  - Avoid short-acting CCBs (e.g. nifedipine IR) → risk of reflex tachycardia
- Stable angina: CCB can be used alone or combined with LA-nitrate or beta blocker
  - Non-DHP CCB + BB is not recommended → risk of bradycardia and left ventricular dysfunction

PHARMACOLOGICAL MEASURES - CCBS

Non-DHP → diltiazem, verapamil; DHP → amlodipine, nifedipine ER

Stable angina: both non-DHP and DHP CCBs can be used alone or combined with LA-nitrate or *beta blocker

- *Note: Non-DHP CCB + BB is not recommended due to risk of bradycardia and AV block

ACS: switch from CCB-DHP to BB if possible

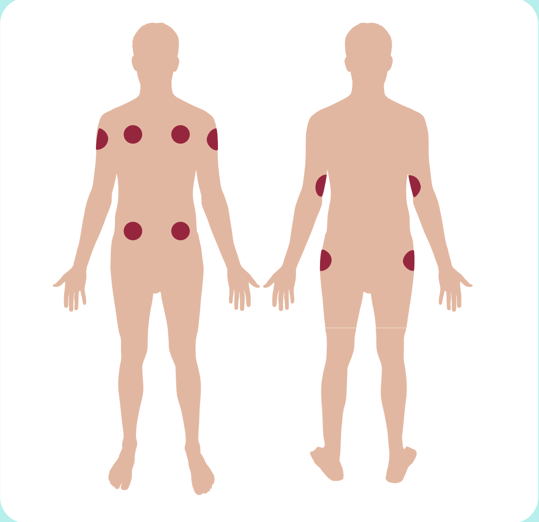
Unstable angina/NSTEMI	STEMI
Use in patients receiving maximally tolerated dose of beta-blockers and adequate doses of nitrates & in patients when BB is contraindicated Caution: Non-DHP CCB (e.g. diltiazem, verapamil) → risk of avoid left ventricular (LV) dysfunction, severe bradycardia or increasing atrioventricular (AV) nodal block Avoid: Avoid short-acting CCBs (e.g. nifedipine IR) → acute drop in blood pressure increases risk of stroke (reflex tachycardia)	Not recommended due to increase in morbidity and mortality If patients have atrial fibrillation and beta blockers are contraindicated, can consider use cautiously to relieve ischemia or to achieve rate control Consider low-dose diltiazem, e.g., 90–120 mg daily in divided doses, with heart rate monitored closely

PHARMACOLOGICAL MEASURES – NITRATES

Acute Coronary Syndromes (NSTEMI, STEMI, Unstable Angina)	Stable Angina
• Short-acting (SL spray, SL tablets, IV) → for acute symptom relief • No mortality benefit • Drug interactions: PDE-5 inhibitors (e.g. sildenafil, tadalafil, vardenafil) → severe hypotension • Side effects: headaches, syncope	• Improve quality of life, no mortality benefit • Short-acting (SL spray, SL tablets) → for acute symptom relief • Long-acting (nitroglycerin transdermal patch, isosorbide dinitrate PO tablets) → used chronically to prevent symptoms - 10-12 hour/day nitrate-free period important for preventing tolerance • Drug interactions: PDE-5 inhibitors (e.g. sildenafil, tadalafil, vardenafil) → severe hypotension • Side effects: headaches, syncope

PHARMACOLOGICAL MEASURES - NITRATES

Nitroglycerin SL spray	• Priming → before first use, and if not used for >14 days • DO NOT SHAKE (creates air bubbles → can result in delivery of an incomplete dose) • Spray onto or under tongue, administer at 5-minute intervals for up to 3 doses • Sit down before administration to prevent syncope
Nitroglycerin patch	• Apply patch QAM and remove QHS • Recommended application sites include upper arms or chest • Rotate application sites • Do NOT apply below elbows or knees



Nitroglycerin patch

MONITORING PARAMETERS

Efficacy Endpoints

Parameter	Degree of Change	Timeframe
Chest pain/pressure	Improved	Hours to days
BP	SBP < 130 mmHg	Antihypertensives: 2-4 weeks Lifestyle and dietary factors: 3-4 months
LDL	< 1.8 mmol/L	4-6 weeks
Heart Rate	Resting ~ 60-70 bpm, Exercise < 110 bpm Mean HR < 100 bpm (Holter)	Days

Safety Endpoints

Parameter	Degree of Change	Timeframe
ADR from medication therapy	Prevent	Duration of therapy
Recurrent symptoms	Prevent	Duration of therapy

TAKEAWAYS

Stable Angina: exercise-induced; no/minimal ECG changes with no changes in troponin

- Symptom management
  - Acute: NTG SL tabs or spray Q5MIN PRN
  - Chronic: BB, LA-CCB, or LA-Nitrate (when BB or LA-CCB cannot be used)
  - Caution: BB + non-DHP CCB (e.g. diltiazem) → risk of bradycardia, AV block, excessive fatigue
- Decrease mortality
  - Start: antiplatelet (ASA 81 mg daily or clopidogrel 75 mg daily if intolerant to ASA), high-intensity statin, and ACEi or ARB
  - Stop: HRT, NSAIDs

Unstable angina: often occurs at rest; ST depression with no changes in troponin; may lead to heart attack

Myocardial infarction: ST-Elevation = STEMI, ST-depression = NSTEMI

- UA/NSTEMI/STEMI: ASA STAT, O_2 PRN (maintain O2 >90%), morphine PRN, NTG IV/SL (unless SBP <90), BB (if CI: CCB), heparin
- UA/NSTEMI: High risk → PCI and continue ASA + P2Y12 inhibitor, optimize antianginals
  - Not high risk: Continue ASA, P2Y12 inhibitor, heparin, BB, start ACEi within 24 hours, high-potency statin, optimize antianginals
- STEMI: PCI is preferred if → PCI facility available and can be performed in ≤ 120 min (door-to-needle)
  - If PCI not feasible → Fibrinolysis +/- PCI and ACEi (start within 24 hours post-MI), ASA 81 mg PO daily continued lifelong unless CI, clopidogrel 75 mg PO daily – DAPT x 1 year, heparin, enoxaparin or fondaparinux (minimum 48 hrs OR until PCI underway), DVT prophylaxis until ambulatory (if not on anticoagulant)
- POST STEMI/NSTEMI Drug Therapy: DAPT, high-intensity statin, BB, ACEi

CASE 1

QB is a 66-year-old male presenting to the ER with sudden-onset midsternal chest pain which started while he was shoveling his driveway. As previously instructed by his family doctor, QB took his nitroglycerin spray at the onset of pain, then every 5 minutes for a total of 3 times. There was no relief or improvement in his chest pain. QB has type 2 diabetes for which he takes gliclazide MR 120 mg PO daily and metformin 1000 mg PO BID. QB admits that he forgets to take his medications occasionally. QB is a high-school math teacher. He does not smoke, but drinks 4-5 alcoholic beverages per week. He doesn't use any recreational drugs.

Vital signs: BP 129/76 mmHg, HR 93 bpm, O_2 97% on room air
ECG: ST segment depression in V3 and V4
Labs: Na 142 mmol/L, LDL 2.6 mmol/L, CRP 15 mg/L, Troponin I: 25 ng/L (normal ≤ 45 ng/L), A1C: 6.9%

1. Which of the following is the most likely cause of QB's symptoms?

   a) STEMI
   b) NSTEMI
   c) Unstable Angina
   d) Stable Angina

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2. Which of the following is not one of QB's risk factors for ACS?

a) Age
b) Diabetes Mellitus
c) Gender
d) Physical activity

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3. All the following should be administered to QB immediately EXCEPT:

   a) ASA 160 mg
   b) Metoprolol 25 mg
   c) Enoxaparin 1 mg/kg
   d) Prasugrel 10 mg daily

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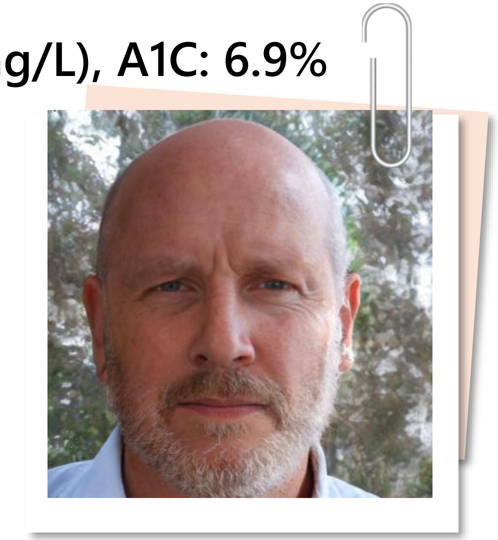

4. Which of the following is NOT an initial treatment option for QB?

a) Medical management
b) PCI
c) CABG
d) Fibrinolysis

CASE 1

QB is a 66-year-old male presenting to the ER with sudden-onset midsternal chest pain which started while he was shoveling his driveway. As previously instructed by his family doctor, QB took his nitroglycerin spray at the onset of pain, then every 5 minutes for a total of 3 times. There was no relief or improvement in his chest pain. QB has type 2 diabetes and stable angina, for which he takes gliclazide MR 120 mg PO daily, metformin 1000 mg PO BID and nitroglycerin spray 0.4 mg every 5 minutes PRN. QB admits that he forgets to take his medications occasionally. QB is a high-school math teacher. He does not smoke, but drinks 4-5 alcoholic beverages per week.

Vital signs: BP 129/76 mmHg, HR 93 bpm, O_2 97% on room air
ECG: ST segment depression in V3 and V4
Labs: Na 142 mmol/L, LDL 2.6 mmol/L, CRP 15 mg/L, Troponin I: 25 ng/L (normal ≤ 45 ng/L), A1C: 6.9%

5. Upon discharge, which of the following medications should be prescribed for QB?

a) Pravastatin 40 mg PO daily
b) Diltiazem XC 240 mg PO QHS
c) Bisoprolol 5 mg PO daily
d) ASA 650 mg PO QID

CASE 2

SM is a 72-year-old male presenting to your pre-admission clinic for an upcoming elective CABG in 3 weeks. He is quite anxious about the procedure and has several questions about his medications. SM has a medical history of chronic stable angina grade 3, dyslipidemia, hypertension, benign prostatic hyperplasia (BPH) and transient ischemic attack (TIA). His pharmacy profile shows he takes clopidogrel 75 mg PO daily, ISDN 10 mg PO TID, valsartan 160 mg PO daily, and pravastatin 40 mg PO QHS. SM does not smoke or use recreational drugs. SM used to drink approximately 2-3 drinks per day but quit 5 years ago.

1. Which of the following medications should be stopped, and when, before CABG?

   a) ISDN 5 days prior to CABG
   b) Valsartan 3 day prior to CABG
   c) Pravastatin 7 days before CABG
   d) Clopidogrel 5 days before CABG

CASE 2

SM is a 72-year-old male presenting to your pre-admission clinic for an upcoming elective CABG in 3 weeks. He is quite anxious about the procedure and has several questions about his medications. SM has a medical history of chronic stable angina grade 3, dyslipidemia, hypertension, benign prostatic hyperplasia (BPH) and transient ischemic attack (TIA). His pharmacy profile shows he takes clopidogrel 75 mg PO daily, ISDN 10 mg PO TID, valsartan 160 mg PO daily, and pravastatin 40 mg PO QHS. SM does not smoke or use recreational drugs. SM used to drink approximately 2-3 drinks per day but quit 5 years ago.

2. What therapy can be added to SM's regimen to improve his anginal symptoms?

   a) Atenolol 25 mg PO daily
   b) ASA 81 mg PO daily
   c) Terazosin 5 mg PO QHS
   d) HCTZ 12.5 mg PO Daily

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CHANGE LOG

March 2025

- Slides 16 and 18: Added names of fibrinolytic agents

July 2025

- Minor formatting edits made
- Slide 22: Changed duration to hold ticagrelor to 2-3 days
- Slide 23: Noted reduced ticagrelor dose

September 2025

- Formatting changes made throughout, re-arranged some slides for better flow
  Slides 7 &10: Added side note on ECG variations of NSETMI and UA
- Slide 12: Added ‘symptom management’ to algorithm
- Slide 15: Added high-potency statin to algorithm
- Added details to cases 1 & 2
- Slide 27: Changed BP target to SBP < 130 mmHg

January 2026

- Slide 2: Updated NAPRA competencies slide
- Slide 8: Added slide to elaborate on troponin vs. creatine kinase – more for FYI
- Slide 13: Added that beta blockers have mortality benefits in specific populations
- Slide 21: Added min # of days to stop antiplatelets before a semi-urgent vs elective CABG surgery
- Slide 24: Added CI of BB